

The Cliff That Wasn't: How Numerical Tolerance Miscalibration Manufactured a Capacity Collapse (and What Caught It)

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Abstract

A wide-grid scaling sweep on a trained matrix fast-weight memory appeared to show a capacity collapse: at state dimension $d_{\text{state}} = 96$, 11 of 12 new load cells ($K \in \{72, 78, 84, 90\}$) failed an eval-time admission gate – a result that, reported as measured, would have been a headline negative finding, directly contradicting the same architecture’s clean super-linear capacity growth already measured from $d_{\text{state}} = 64$ to $d_{\text{state}} = 80$. We report the diagnosis instead. The admission gate’s Newton–Schulz convergence check, tuned against anchor-stabilized *blended* keys, was miscalibrated for a structurally different, un-stabilized *raw* key population that one specific evaluation probe pool actually queries. A purpose-built, four-times independently-audited repro instrument recovered the true failing tensor and showed every one of 4,608 flagged episodes resolves cleanly under a modestly larger iteration budget ($n_{\text{iter}} = 20 \rightarrow 28$) – a near-miss, not a structural wall. Recalibrating the admission *flag* alone (never the underlying capacity measurement) unlocked the quarantined cells at zero additional GPU spend; the re-fit shows no cliff through $K/d_{\text{state}} = 0.9375$. We report four further findings from the same instrument-audit chain: the recalibrated frontier itself moves with load (sufficient at $K=84$, insufficient at $K=90$); the failing admission leg swaps identity across recalibrations; an archived exact-ceiling reading did not replicate at a fresh seed; and a diagnosed confound shows a real, directionally-consistent effect roughly twelve times its own measured noise floor, correctly withheld from promotion to a finding because its own diagnostic cell never cleared admission. We describe the discipline – pre-registered decision rules over instrument state, independent adversarial audit rounds with fresh eyes at each round, negative unit tests exercised (and mutated) to completion, and verification against raw artifacts rather than summary logs – that caught the miscalibration before it became a published claim, and argue it generalizes to any measurement-heavy empirical ML claim gated by a numerical admission check.

1 Introduction

Empirical claims about learned systems are only as trustworthy as the instrument that measured them. This is a case study of one instrument failing quietly, being caught before publication, and being fixed – reported because the failure mode (a numerical tolerance calibrated against one data population, silently misapplied to a structurally different one) and the practice that caught it (pre-registered decision rules, independent adversarial audit, verification against raw artifacts) generalize to any measurement-heavy ML claim gated by a numerical admission check. **The setup, one sentence:** a trained matrix fast-weight memory’s associative-recall capacity is measured by exact continuous recovery across a load sweep K/d_{state} , and a separate, eval-time admission gate decides which measured cells are trustworthy enough to enter a capacity fit.

Scope and companion paper. A companion paper [5] reports this program’s capacity *results*: the cliff located at $d_{\text{state}} = 64$, its dissolution at $d_{\text{state}} = 128$, and a coherence confound tested and exonerated; its own Limitations section discloses several of the same raw numbers analyzed here, as scope-limited caveats on its $d_{\text{state}} = 96$ material. This paper’s claim is different in kind: it is about the measurement *failure mode itself* – how a capacity collapse was manufactured by an uncalibrated instrument, and what process caught it – not a new capacity-boundary result. Where a number appears in both papers, this paper cites it for what the audit trail says about instrument reliability, not to re-claim it as a located transition.

2 Setup

The testbed is a single-head, single-layer DeltaNet-style fast-weight memory [8]: a per-entity key-anchoring mechanism provides a table of 107 trainable entity keys, and capacity is measured by held-out compositional recovery at hop depth 4 ($h4$), graded by *exact continuous* cosine similarity against the true target vector at threshold 0.9 – never argmax over a codebook, the design choice that keeps a rank or capacity bound from being silently satisfiable by codebook lookup [7]. K is the number of independent entity bindings one episode must simultaneously support; K/d_{state} is the capacity ratio the load sweeps below vary.

Not every trained cell’s $h4$ is automatically eligible for a capacity fit: an eval-time *admission gate* must clear four disjoint legs – a value-salvage tier check, a finite-loss/no-divergence check, a task-performance floor, and a Newton–Schulz convergence check – before a cell enters a fit.¹ The leg this paper is about, Newton–Schulz convergence (`ns_converged_no_fallback`), gates whether an orthogonalization procedure – an ancillary computation *inside the recovery-probe scoring path*, not part of the trained model itself – converges to a residual under `resid_tol = 0.01` within a fixed iteration budget n_{iter} (production setting $n_{\text{iter}} = 20$).

3 The Phenomenon: An Apparent Capacity Collapse at $d_{\text{state}} = 96$

A wide-grid load sweep at $d_{\text{state}} = 96$ ran 12 new cells ($K \in \{72, 78, 84, 90\}$, 3 seeds each) alongside 3 reused $K=69$ cells. Training was clean at every single cell – $h1$ (the in-distribution training-health guard) read exactly 1.0 at all 19, and $n_{\text{geo3_fallback_train_steps}} = 0$ everywhere – but of the 12 new cells, only **1 was admissible**: $K=78$, $K=84$, and $K=90$ each failed admission on all 3 of their own seeds (0/3 each), an overall new-cell admissibility rate of **5/16 (31.25%)**. Reported as measured, this is the shape of a genuine capacity cliff: a sharp admissibility collapse rising from 0/30 at $K/d_{\text{state}} \leq 0.71875$ to 3/3 inadmissible at every $K/d_{\text{state}} \geq 0.8125$ – directly contradicting the clean, non-degenerate super-linear growth already measured between $d_{\text{state}} = 64$ and $d_{\text{state}} = 80$ ($x_0 = 0.5455 \rightarrow 0.6779$, 5): a newsworthy, and as it turned out wrong, negative result.

The *first* mechanistic hypothesis was itself wrong, and catching that was the episode’s first lesson. The initial writeup attributed the failure to the trained anchor table’s own Newton–Schulz convergence at high K/d_{state} . An independent follow-up diagnostic tested that claim directly on six pulled checkpoints (four failing, two passing controls) and refuted it: the anchor table converges cleanly at $n_{\text{iter}} = 20$ with residuals $\sim 7,000\text{--}8,000\times$ **below** the 0.01 tolerance, indistinguishable between failing and passing cells. The real failure traced to a single evaluation probe pool (`C17_heldout_entities`) that is, by construction, architecturally anchor-*bypassed* – the anchor table’s own conditioning cannot input a computation it never participates in. The original mechanistic claim was formally retracted before any fix was attempted.

4 The Catch: Instrument Saturation, Not Model Failure

With the wrong object identified and ruled out, a purpose-built repro instrument was designed to recover the *true* failing tensor: C17’s raw, pre-Newton–Schulz post-conv keys for held-out entities, which the original wave’s checkpoint writer never saved. The design went through **four independent adversarial attack rounds** before any GPU cell launched, each catching real defects, not merely stylistic ones. Round 1’s FATAL fix closed a gap where an empty telemetry dump (a plausible outcome, given this hardware’s own measured fixed-seed run-to-run nondeterminism) would let a universally quantified check vacuously pass and silently emit an unearned TOLERANCE- MISCALIBRATION verdict – in the design’s own words, “an unearned ‘unlock the 11 cells’ paper claim.” Round 4’s MAJOR fix closed a “two-seeds trap”: the offline analysis reconstructs the training/held-out entity partition from a *hardcoded* seed decoupled from the cell’s own launch seed – naively threading the launch seed in would silently reconstruct the wrong partition and produce a confidently wrong verdict on healthy code.

The instrument’s own decision rule was pre-registered as a strict precedence order over telemetry state, walked mechanically at harvest time: a pool-membership precheck (any single violation is dispositive, structural, no statistics required) $\rightarrow$ a reproduction-count floor $\rightarrow$ live/offline Newton–Schulz agreement $\rightarrow$ an n_{iter} sweep. Every step ran, in order, on the real cell ($K=84$, seed= 1940): the reconstruction gate passed (107/107 set-equal); **zero** pool-membership violations across all 36 dumped events (ruling out an instrument bug); the reproduction floor cleared at

¹Exact identifiers, verified against the live code: `value_salvage_tier_pass`, `finite_loss_no_divergence`, `task_performance_floor_pass`, `ns_converged_no_fallback`.

12 $\times$ its own minimum; **zero** live/offline disagreements across all **4,608** examined episodes (ruling out an instrument bug a second, independent way); and the n_{iter} sweep resolved **every single one** of those 4,608 episodes by $n_{\text{iter}} \leq 28$ (4,313 at $n_{\text{iter}} = 24$, the remaining 295 at $n_{\text{iter}} = 28$, zero left unresolved even at $n_{\text{iter}} = 32$ or 40) $\rightarrow$ TOLERANCE-MISCALIBRATION.

The magnitude makes the "near-miss, not a wall" reading concrete: at the final checkpoint, the worst-case flagged episode's residuals ranged **0.696–1.371** against the 0.01 tolerance – 70 $\times$ –137 $\times$ over threshold – and this near-miss population *worsened monotonically* over training (13–27 of 128 anomalous rows at step 16,000, all 128/128 by step 20,000). Yet it still cleared cleanly by $n_{\text{iter}} = 28$. A structural wall would be expected to keep worsening past $n_{\text{iter}} = 40$; a miscalibrated tolerance resolves at a fixed, modest iteration bump, and did.

Because the underlying capacity measurement was never in question – training converged cleanly at all 11 quarantined cells regardless of admission status – only the admission *flag*, never *h4*, was recalibrated, and only for the 11 cells independently verified to share the identical failure signature (one pre-existing, out-of-scope anomaly at $K=69$ was deliberately *not* flipped, a disclosed scope decision, not an oversight). This unlocked the full grid at **zero additional GPU spend**. The re-fit found no cliff anywhere through $K/d_{\text{state}} = 0.9375$: per- K mean *h4* reads 0.9592($K=69$), 0.9216($K=72$), 0.9326($K=78$), 0.9581($K=84$), 1.0000($K=90$) – non-monotonic, but nowhere collapsing.

5 General Findings: Admission Checks Are Instrument-Relative

The follow-up diagnostic chain (a zero-GPU power check, then a two-cell control diagnostic at the recalibrated $n_{\text{iter}} = 28$) surfaced five findings that generalize beyond this one instrument.

(a) **The n_{iter} -sufficiency frontier moves with load.** $n_{\text{iter}} = 28$ resolved every flagged episode at $K=84$ (§4). A follow-up control cell at $K=90$, run under the *same* recalibrated $n_{\text{iter}} = 28$, stayed fallback-inadmissible: a frontier measured at one K/d_{state} does not transfer to a higher one without re-verification.

(b) **Admission legs swap failure modes across recalibrations.** At $n_{\text{iter}} = 28$, $K=84$ fails a *different* leg (*value_salvage_tier_pass*, $0.09307 < 0.10$, training convergence now fully clean) than the original $n_{\text{iter}} = 20$ population, which failed convergence while *value-salvage passed* (0.102–0.126). Fixing one failure mode can silently expose the next.

(c) **An archived exact ceiling did not replicate.** $K=90$'s original 3 seeds all read exact *h4* = 1.0000; a fresh seed at the recalibrated $n_{\text{iter}} = 28$ read 0.9725 – "exact" described the specific seeds measured, not a property of the load.

(d) **A diagnosed confound is real in direction but was correctly withheld.** Restricting $K=84$'s eval-time entity-pool margin to match $K=90$'s own overlap fraction raised *h4* by +0.0330 – roughly **12 $\times$** the measured eval-sampling noise floor (0.0028, the larger of two independent noise draws). The mechanism is directionally real, but because *both* of this diagnostic's own cells stayed inadmissible even at $n_{\text{iter}} = 28$, the pre-registered routing correctly refused to promote it to a finding – the discipline that caught the original miscalibration also declined to over-claim its own follow-up.

(e) **Structural checks need exact thresholds, never tolerance slack borrowed from a numerical context.** This instrument's own attack history supplies the counter-example: an early draft of the pool-membership precheck reused a ≥ 2 -event recurrence floor built for a genuinely noisy *numerical* disagreement check, applied unexamined to a *structural* fact – pool membership has no floating-point component; an id either is or is not in a disjoint set. Under the reused floor, a single real violation in a small event set would be silently excluded, and the verdict would continue, confidently and wrongly, toward TOLERANCE-MISCALIBRATION. The fix split the floor: a single structural violation is dispositive on its own; the recurrence floor is retained only for the numerical check it was designed for.

6 The Methodology That Caught It

Table 1 names six waves; four independent adversarial audit rounds landed on the repro instrument's own design alone (§4), and a fifth wave's design took two further rounds plus a dedicated verification pass before any GPU cell launched. That later rounds keep finding real defects – round 4's two-seeds trap (§4) is not a first-round-only phenomenon – is itself the argument against stopping at one audit pass.

This one case study sits inside a well-documented broader pattern: reported ML results are known to be sensitive to analytic and implementation choices invisible in the final number [2, 10], that empirical rigor has not obviously kept

§	Wave	Verdict
15.22	$d_{state}=96$ wide-grid load sweep	AMBIGUOUS (data-quality collapse)
15.23	anchor-table NS diagnostic	MISDIAGNOSED-ARTIFACT
15.24	C17 repro instrument, design	DESIGN-CLEARED (4 audit rounds)
15.25	C17 repro instrument, harvest	TOLERANCE-MISCALIBRATION → unlock → re-fit AMBIGUOUS (non-monotonic, no cliff)
15.26	scatter-resolution power check	finding registered (zero GPU); 10-cell grid killed
15.27	K90 pool-margin control	DEGENERATE_CELL (3 escalated findings, PI-gated)

Table 1: The full diagnostic chain and its verdicts, each anchored to its own § in `KEY_ANCHORING_SCALING_DRAFT.md`. No step skipped or silently absorbed a prior step’s finding.

pace with the field’s own pace of progress [6, 9], and that unaccounted sources of variance routinely change which of two systems looks better [1, 3]; reporting-standard proposals such as REFORMS [4] formalize much of the raw-artifact verification and pre-registration this instrument-audit chain practiced informally, ahead of any such checklist.

Four practices recur across every wave in Table 1 and are offered here as transferable, not architecture-specific:

- **Pre-registration of outcome semantics.** The repro instrument’s six-step decision rule mapped every possible telemetry state to exactly one named verdict, fixed *before* the cell that would decide this paper’s own headline claim ran.
- **Negative tests exercised, and mutated, to completion.** A synthetic empty-sink fixture had to emit NO-REPRO, never a verdict; a synthetic single-violation pool-mismatch fixture had to emit INSTRUMENT-BUG unconditionally; the pool-margin wave’s own safety guards were *mutation-tested* – reverting a guard had to make its own regression suite fail, not merely pass when correct.
- **Spend compute only where it changes the answer.** A registered 10-cell, 4.27–8.54 GPU-h seed-escalation sweep was killed and replaced by a 360,000-trial, zero-GPU bootstrap power check (8 seeds, an independent reimplement, a positive control proving the check has teeth) that answered the same question at CPU cost.
- **Verify against raw artifacts, not summary logs.** Every headline number here – the admissibility table, the 4,608-episode n_{iter} -sweep tally, the pool-margin shift and noise floor – was independently re-derived, at the harvest that reported it, from raw JSON/.pbt dumps or a fresh read-only box pull, not re-read from an earlier commit message.

The economics are worth stating plainly: the repro instrument that identified TOLERANCE-MISCALIBRATION cost **0.82 GPU-h**; the follow-up pool-margin diagnostic cost **1.09 GPU-h** – under **2 GPU-h combined** to interrogate an instrument whose uncorrected reading would have discredited **6.33 GPU-h** of clean training compute already spent, and shipped as a false 11-of-12-cell capacity collapse. Auditing the instrument was cheap relative to the compute it was protecting.

7 Limitations

Single architecture family. All findings come from one DeltaNet-style fast-weight testbed and one specific admission gate (Newton–Schulz residual tolerance inside an eval-time recovery probe). No claim is made that this exact failure mode, or this exact fix, recurs in other admission gates, architectures, or codebases – only that this one, checked, was miscalibrated and is now fixed.

The ceiling fine structure remains genuinely open. Finding (d)’s pool-margin mechanism is real in direction but the diagnostic built to confirm or refute it landed DEGENERATE_CELL: neither of its own two cells cleared admission even at $n_{iter} = 28$, so its own adjudication triggers never fired. This is registered for a follow-up design round and is explicitly **PI-gated**, not run as of this writing.

One instrument, one recalibration, no counterfactual. This paper documents one class of numerical miscalibration (Newton–Schulz residual tolerance, calibrated against a blended-key population, silently misapplied to a raw-key one) and one recalibration of it – not a general certificate that every admission gate in this codebase, or any other, is now trustworthy. The methodology narrative is reconstructed from this project’s own audit trail, not a controlled comparison against a matched project that skipped the audit steps; we cannot report a rate of miscalibration a lighter process would have missed, only that this process caught the one it did.

Companion-paper boundary, restated. The capacity numbers this paper cites for provenance (x_0 progression, the $d_{\text{state}} = 96$ re-fit, the pool-margin diagnostic’s own descriptive readings) are also reported, as scope-limited caveats, in Larson [5]. That paper’s claim is about the located transition and its dissolution with scale; this paper’s claim is about the measurement failure mode and the process that caught it before it became one.

References

- [1] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, 2021. arXiv:2103.03098.
- [2] Alexander D’Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022. arXiv:2011.03395.
- [3] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018. arXiv:1709.06560.
- [4] Sayash Kapoor, Emily Cantrell, Kenny Peng, Thanh Hien Pham, Christopher A. Bail, Odd Erik Gundersen, Jake M. Hofman, Jessica Hullman, Michael A. Lones, Momin M. Malik, Priyanka Nanayakkara, Russell A. Poldrack, Inioluwa Deborah Raji, Michael Roberts, Matthew J. Salganik, Marta Serra-Garcia, Brandon M. Stewart, Gilles Vandewiele, and Arvind Narayanan. REFORMS: Reporting standards for machine learning based science. *arXiv preprint arXiv:2308.07832*, 2023.
- [5] Sam Larson. The capacity cliff is not capacity: Locating, dissolving, and exonerating a coherence confound in trained fast-weight memories. Companion paper, draft, venue pending PI decision (matrix-thinking/submissions/workshop-2026/). Reports the substantive capacity results this paper’s case study is drawn from; this paper reports the measurement failure mode and audit process, not the capacity results themselves., 2026.
- [6] Zachary C. Lipton and Jacob Steinhardt. Troubling trends in machine learning scholarship. In *ICML 2018 Machine Learning Debates Workshop*, 2018. arXiv:1807.03341.
- [7] Eshaan Nichani, Jason D. Lee, and Alberto Bietti. Understanding factual recall in transformers via associative memories. In *International Conference on Learning Representations (ICLR), Spotlight*, 2025. arXiv:2412.06538.
- [8] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, 2021. arXiv:2102.11174.
- [9] D. Sculley, Jasper Snoek, Alex Wiltschko, and Ali Rahimi. Winner’s curse? on pace, progress, and empirical rigor. In *International Conference on Learning Representations (ICLR) Workshop Track*, 2018.
- [10] Raphael Silberzahn, Eric Luis Uhlmann, Daniel P. Martin, Pasquale Anselmi, Frederik Aust, Eli Awtrey, Štěpán Bahník, Feng Bai, Colin Bannard, Evelina Bonnier, et al. Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science*, 1(3): 337–356, 2018.